Labelling using the annotate() method We can display and mark several mathematical functions in a plot using **annotate()** method. Letus understand this using a problem: Problem: Plot and label resistance characteristics and a power hyperbola Write a Python program using NumPy and Matplotlib to plot the following functions on the same graph:

1. Three resistance characteristic curves:

   U1=R1 I,U2=R2 I, U3=R3 I

   where

   R1=2 Omega, R2=4 Omega, R3=8 Omega

2. A power hyperbola:

   U=P/I

   P=1W

3. Generate current values I between 0.1 A and 1 A using np.linspace().
4. Plot the three resistance characteristics and the power hyperbola on the same set of axes.
5. Use different colours to distinguish the curves.
6. Label the three resistance curves as R1, R2, and R3 using the annotate() method.
7. Add appropriate labels for the x-axis and y-axis, a title, and a grid.

```
import numpy as np
import matplotlib.pyplot as plt

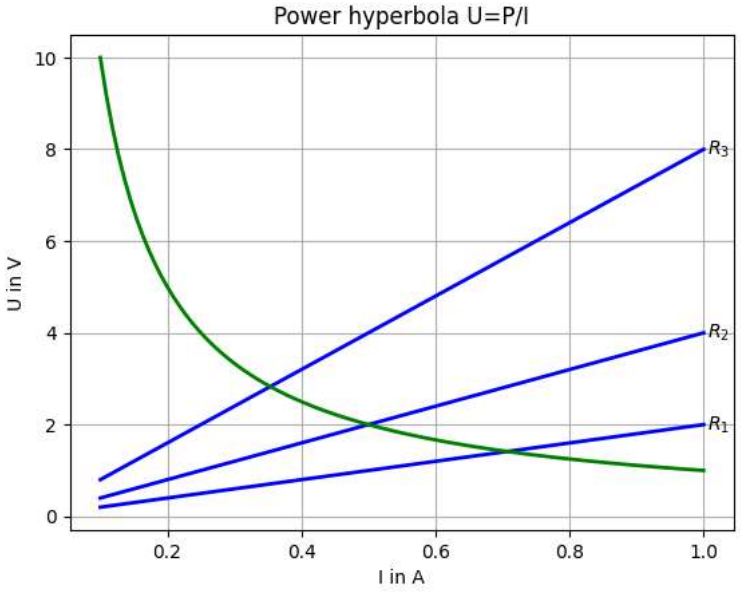
R1,R2,R3=2,4,8
I=0.2
P=1
I=np.linspace(0.1, 1, 100)

U1=R1*I
U2=R2*I
U3=R3*I
U=P/I

fig, ax = plt.subplots()

ax.plot(I,U1,I,U2,I,U3,lw=2,color='blue')
ax.plot(I,U,lw=2,color='green')

ax.set(xlabel='I in A',ylabel='U in V',title='Power hyperbola U=P/I')
ax.annotate(r'$R_1$',xy=(1,2),xytext=(+2,-3),textcoords='offset points')
ax.annotate(r'$R_2$',xy=(1,4),xytext=(+2,-3),textcoords='offset points')
ax.annotate(r'$R_3$',xy=(1,8),xytext=(+2,-3),textcoords='offset points')
ax.grid(True)
plt.show()
```

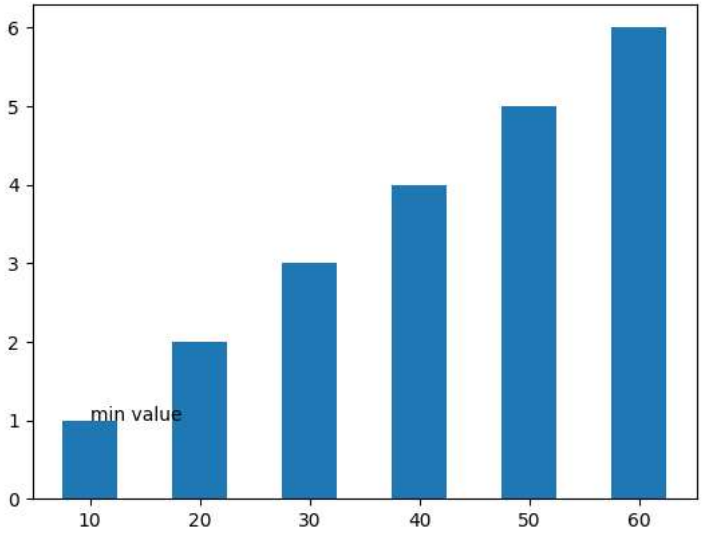



```
import matplotlib.pyplot as plt
x=range(10,61,10)
y=range(1,7)

# data points
print(list(zip(x,y)))
plt.bar(x,y,width=5)
plt.annotate('min value',xy=(10,1))
#plt.annotate('min value',xy=(10,1),horizontalalignment='center',verticalalignment='bottom')

plt.show()
```

```
[(10, 1), (20, 2), (30, 3), (40, 4), (50, 5), (60, 6)]
```




```
import matplotlib.pyplot as plt
x=range(10,61,10)
y=range(1,7)

# data points
print(list(zip(x,y)))
plt.bar(x,y,width=5)

plt.annotate('min value',xy=(10,1),horizontalalignment='center',verticalalignment='bottom')
plt.annotate('Max value',xy=(60,6),horizontalalignment='center',verticalalignment='top')

plt.show()
```

```
[(10, 1), (20, 2), (30, 3), (40, 4), (50, 5), (60, 6)]
```

